Name			
Student ID			
Assessment Venue		Seat Number	

REPUBLIC POLYTECHNIC

E211 Operations Planning II

AY2022 Semester 2 End-Semester Examination (ESE)

Instructions to student:

1) For this question paper, there are **34** pages (including this cover page).
2) Note that textbooks are not allowed. You may refer to paper-based notes (in bound form, and no larger than A4 size) during the assessment.

This segment is to be used by the invigilator only and for 'online' and 'online and paper' mode assessments only.		
Please tick the box below if the student has done part of the assessment online:	Invigilator's Name:	Invigilator's Signature:
☐ Partially done online		

This Segment is to be used by staff grader(s) only.		
Question Number	**Marks Awarded**	**Max Marks**
Section A/Q1		2
Section A/Q2		2
Section A/Q3		2
Section A/Q4		2
Section A/Q5		2
Section A/Q6		17
Section A/Q7		18
Section A/Q8		19
Section A/Q9		13
Section A/Q10		14
Section A/Q11		9
Total		100

Full Marks: 100

Section A (11 Questions) **Maximum Marks for This Section: 100**

Question 1 (2 marks)

Figure 1a shows a drive-through car wash facility. Identify the characteristics of the following queuing system components:
Arrival population, Number of channels, Number of phases

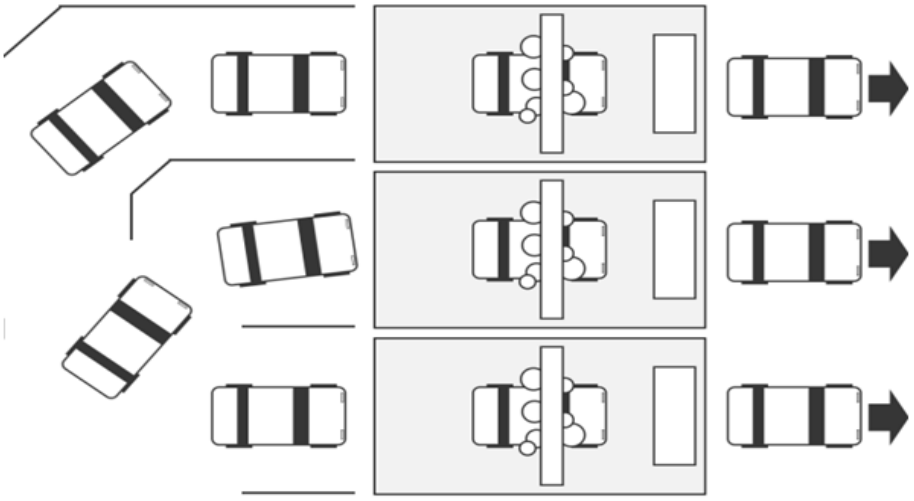

Figure 1a Drive-through Car Wash

(A) Infinite, 3, 3

(B) Infinite, 3, 1

(C) Finite, 3, 3

(D) Poisson, 1, 3

Answer: ( )

Question 2 (2 marks)

Which one of the following objects generates **new** flow items in FlexSim 3D model? (2 marks)

(A) Source

(B) Sink

(C) Queue

(D) Processor

Answer: ( )

Question 3 (2 marks)

In one healthcare facility, patients undergoing a surgery have to be brought to the operating theatre in a gurney (wheeled stretcher). Which HC activity set can be used to depict this process in a FlexSim Healthcare model?

(A) Wait in Line then Process

(B) Escort then Process

(C) Transport then Process

(D) Move Equipment then Process

Answer: ( )

Question 4 (2 marks)

Which method is used to solve a shortest path problem?

(A) Clarke and Wright Savings method

(B) Ford Fulkerson's algorithm

(C) Dijkstra's algorithm

(D) Network analysis

Answer: ( )

Question 5 (2 marks)

Figure 1b shows the price index of HDB resale flats through the past 2 decades. What time series component(s) can be observed in the chart?

Figure 1b Price Index of HDB Resale Flats

(A) Trend and Seasonal

(B) Trend and Cyclical

(C) Seasonal only

(D) Random only

Answer: ( )

Question 6 (17 marks)

Show your workings clearly.
A distribution network shown in Figure 2a distributes automotive parts. Values on each arc represent the unit cost of shipment between 2 locations. Given:

- Plants P1 and P2 can supply **up to** 700 and 500 units respectively
- Warehouses W1 and W2 are transhipment hubs with no demand
- Customers C1 and C2 have demand of 300 and 800 respectively

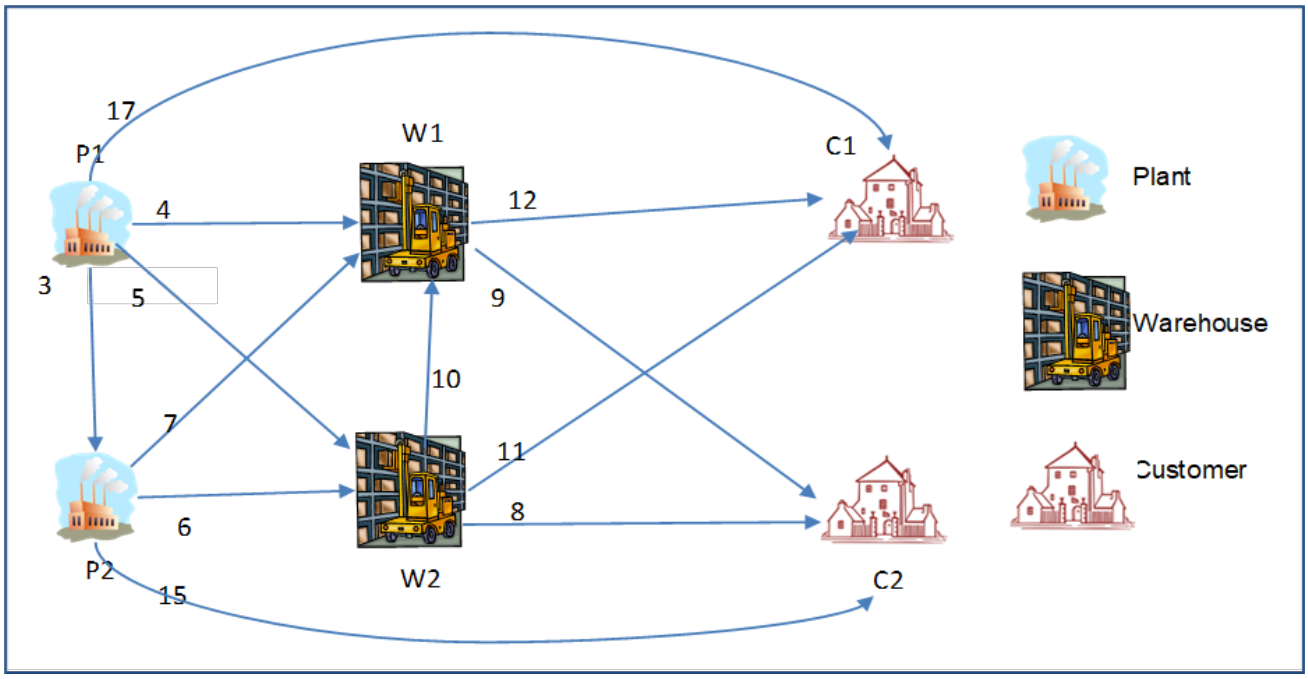

Figure 2a Distribution Network

Question 6a (2 marks)

State in words the objective of formulating a distribution plan to ship parts from Plants to Customers.

Question 6b (6 marks)

Assume Xij (decision variable) represents the quantity of parts to ship from location i to location j, where i, j can be P1, P2, W1, W2, C1 or C2.
Formulate the following constraints:

Question 6b(i) (2 marks)

Net flow constraint for **Plant 1 (P1)**.

Question 6b(ii) (2 marks)

Net flow constraint for **Warehouse 2 (W2)**.

Question 6b(iii) (2 marks)

Flow capacity constraint when at most 150 units can be sent from Plant 2 to Customer 2 directly.

Question 6c (4 marks)

The network diagram in Figure 2b shows the distribution routes for a custom part from Plant 2 to Customer 1. The values on each arc represent the **shipping capacity** along each route. Determine the maximum quantity of the part that can be shipped from Plant 2 to Customer 1, stating clearly the selected routes and shipment amount along each route.

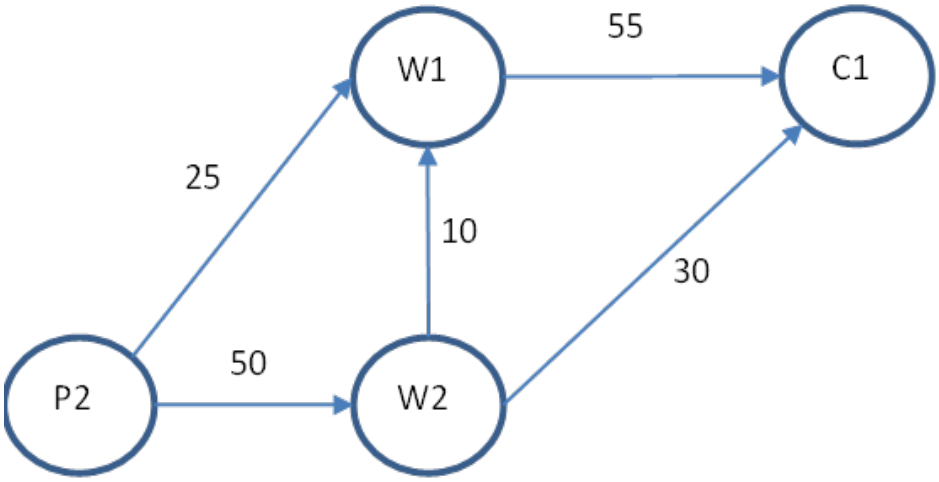

Figure 2b Network Diagram

Question 6d (3 marks)

At one customer site, a part has to be distributed to a number of stores (B, C, D and E). Table 2a shows the travel distances between the locations. The truck will leave the **Depot (location A)**, visit each store once and return to Depot after deliveries. Using the Nearest Neighbour method, work out a delivery route **and** calculate the total travel distance.

Table 2a Travel Distances Between Locations

From	**To (Travel distance in km)**				
	A	**B**	**C**	**D**	**E**
A	-	44	28	36	19
B	44	-	51	31	28
C	28	51	-	20	29
D	36	31	20	-	50
E	19	28	29	50	-

Question 6e (2 marks)

What is another method that can be used for planning delivery routes? Will the method guarantee optimal solutions?

Question 7 (18 marks)

Show your workings clearly**. Round your answer to 2 decimal places where necessary.**
Mei Ling works as a sales manager in a company. Recently she gathered some quarterly sales data of a popular product and plot a time series chart (Figure 3a).

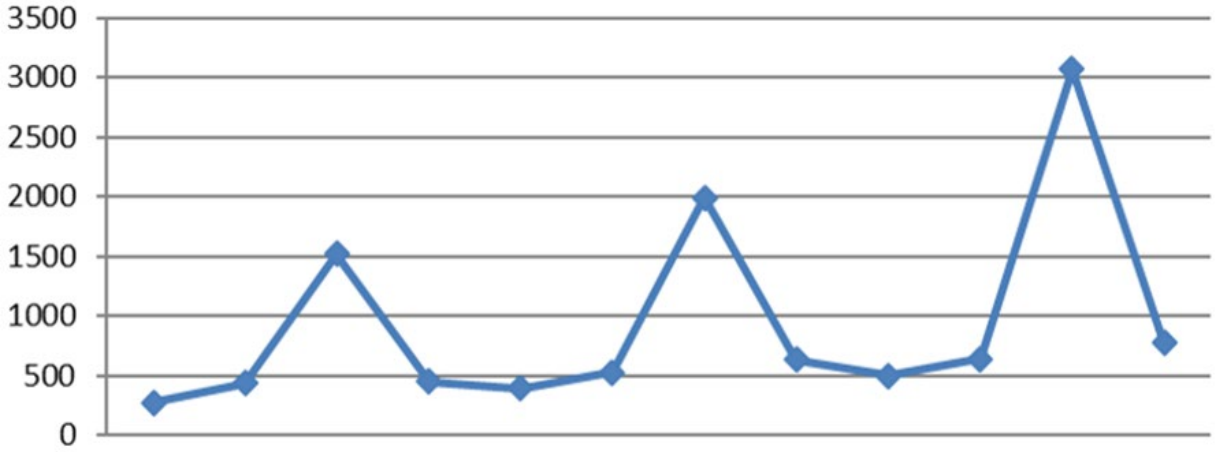

Figure 3a Product Quarterly Sales Figures

Question 7a (2 marks)

Should Mei Ling use Exponential Smoothing method to forecast the sales for the coming quarter? Explain.

Question 7b (2 marks)

To remove the seasonal variation from the data, how many periods of moving average should Mei Ling use? Give a reason. (2 marks)

Question 7c (8 marks)

Refer to the partial workings in Table 3a.

Table 3a Partial Workings of a Forecasting Method

Year/ Quarter	Actual Value (Y)	Time Period (X)	Moving Average	Centred Moving Average	Seasonal Ratio
2021/ 1	410	9	843.75		
2021/ 2	505	10	871.25	'A'	'B'
2021/ 3	1970	11	890		
2021/ 4	600	12	913.75		

Question 7c(i) (2 marks)

Calculate the Centred Moving Average for **2021 Quarter 2** (indicated as 'A' in Table 3a).

Question 7c(ii) (2 marks)

Calculate the Seasonal Ratio for **2021 Quarter 2** (indicated as 'B' in the Table 3a).

Question 7c(iii) (2 marks)

Given that the Seasonal Index for Quarter 2 is 0.61, compute the Deseasonalized Value for **2021 Quarter 2**.

Question 7c(iv) (2 marks)

Given that the linear trend in the deseasonalized values can be represented by a regression model: Y = 260 + 118.2X, compute the sales forecast for **2023 Quarter 2**.

Question 7d (6 marks)

Mei Ling's company recently launched a new product. Table 3b shows the sales numbers in the initial months.

Table 3b New Product Initial Sales

Month	Sales
1	320
2	405
3	375
4	390
5	330

Question 7d(i) (2 marks)

Using Exponential smoothing method, determine the sales forecast for **Month 6**. Assume smoothing constant is 0.3 and forecast for Month 5 is 365.

Question 7d(ii) (2 marks)

Suppose the actual sales figure in Month 6 turns out to be 380. Calculate the absolute percent error of your forecast for Month 6.

Question 7d(iii) (2 marks)

What can Mei Ling do to improve her sales forecast of the new product?

Question 8 (19 marks)

Show your workings clearly and round off your answers to **two decimal places where necessary.** Use ^ symbol to represent the power notation.

Long queues are frequently seen at the packing area of a company's production line. The current packing process is **manual**.

A time study shows that products arrive for packing according to a Poisson process at an average rate of 45 per hour. Average time to pack one product is 1.2 minutes. The manual packing time differs greatly and follows a negative exponential distribution.

The line manager proposes to automate the packing process with a packing machine. The machine packing time is constant at 1.2 minutes per product.

Question 8a (2 marks)

Using Kendall's notation, identify a suitable queuing model for:
i) Manual packing process
ii) Machine packing process

Question 8b (6 marks)

Calculate
i) Average arrival rate of the products to the packing area
ii) Average service rate of manual packing
iii) Utilization of the manual packing process

Question 8c (2 marks)

Calculate the average number of products waiting in queue for the manual packing process, Lq.

Question 8d (2 marks)

Explain in words (without calculations) how the implementation of automated machine packing would affect the average waiting time at the packing area.

Question 8e (3 marks)

Due to a sudden increase in orders, the company must hire additional staff for packing. Determine the minimum number of staff required if product arrival rate is now 150 per hour. Assume average packing time is 1.2 minutes.

Question 8f (2 marks)

With a certain number of staff at the packing area, the average number of products in queue, Lq is 1.53. Calculate the average time each product spent in the system, Ws.

Question 8g (2 marks)

State **two** ways in which a M/M/k queuing model can be used to help **design** the manual packing operations.

Question 9 (13 marks)

Rafi is studying a production line where structural parts are made. The parts first undergo an automated machining process, followed by a Quality Control (QC) check by an operator. He builds a FlexSim process flow simulation model.

Question 9a (6 marks)

Figure 5a shows a snapshot of the model. Answer the following questions:

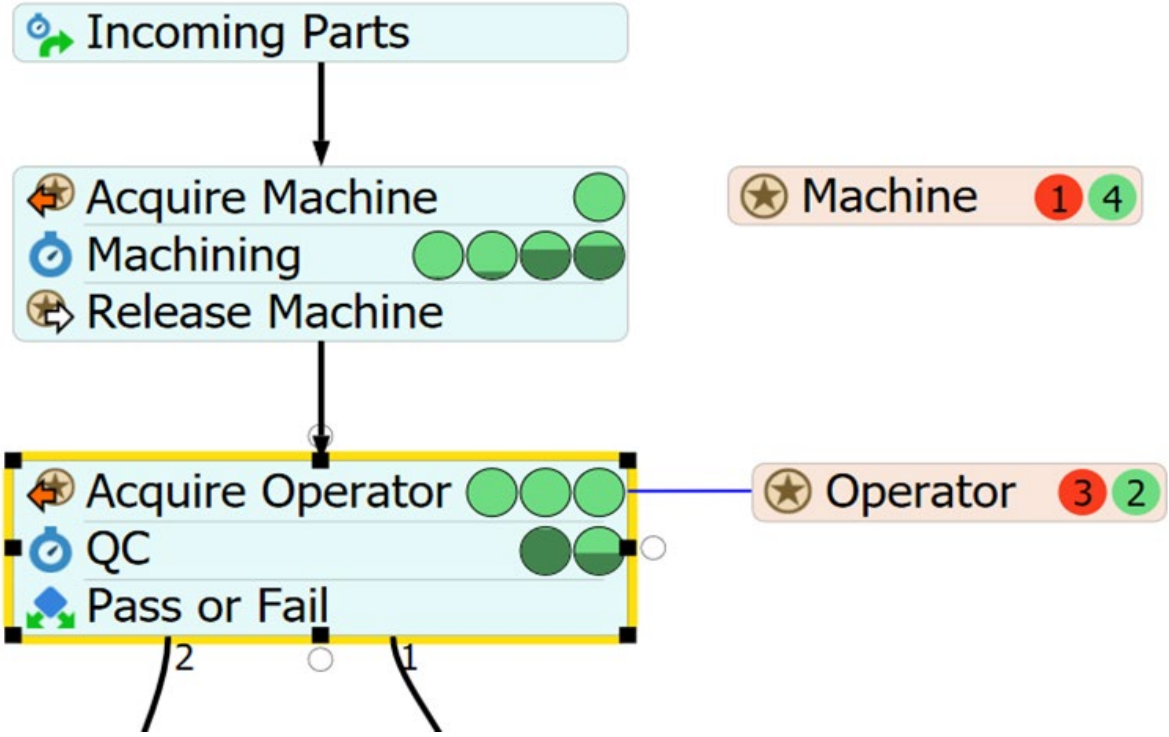

Figure 5a FlexSim Process Flow Simulation

Question 9a(i) (2 marks)

State the number of machines present in the process.

_____

Question 9a(ii) (2 marks)

Suppose there was no 'Release Operator' activity in the model. How would this affect the simulation?

Question 9a(iii) (2 marks)

Name the activity from FlexSim Process Flow library used to model 'Pass or Fail'. What is its function?

Question 9b (5 marks)

Figure 5b shows the statistics of Machine and Operator resources at the end of the simulation. Answer the following questions:

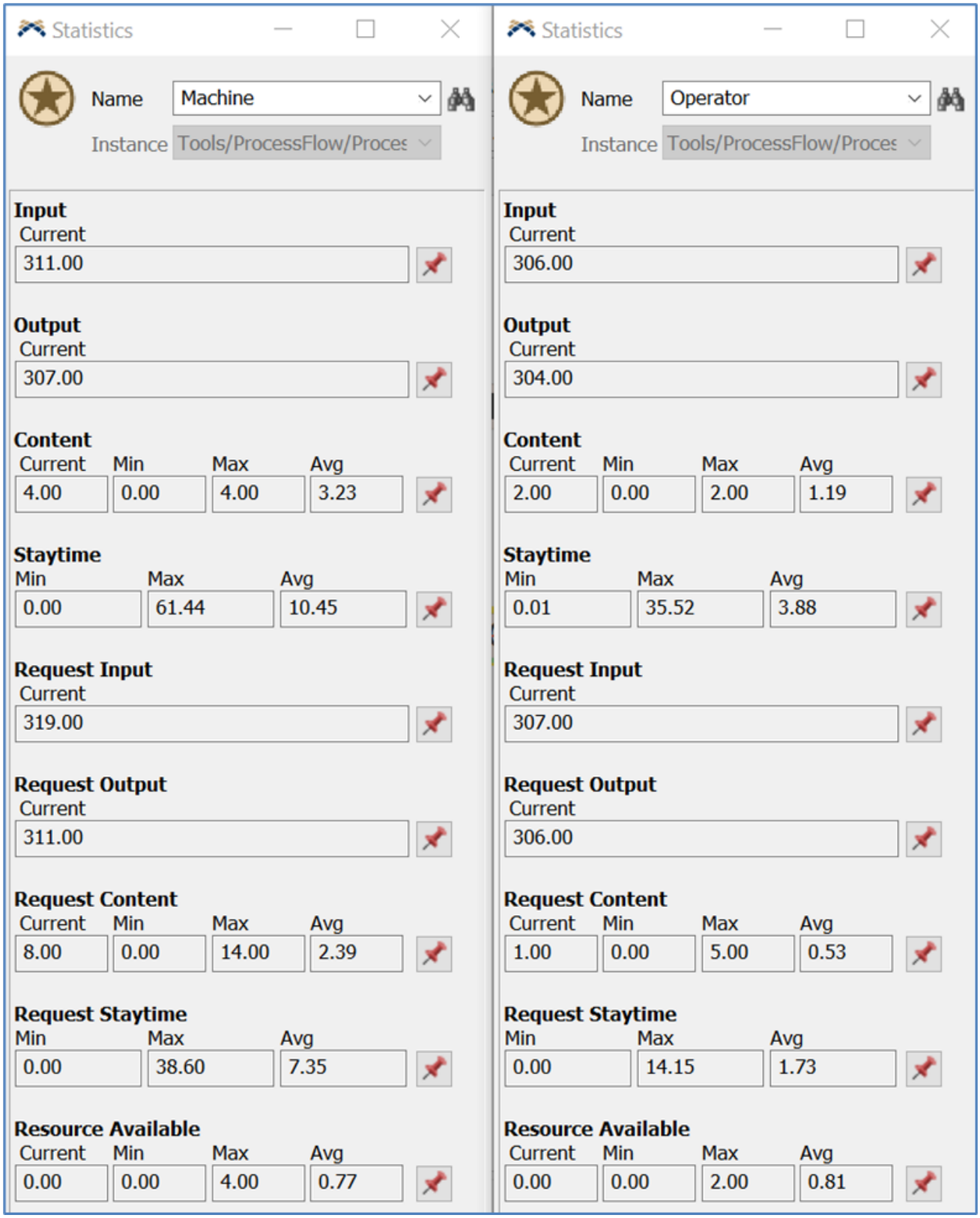

Figure 5b Statistics of Machine and Operator Resources

Question 9b(i) (2 marks)

Determine the bottleneck of the process by comparing the waiting times for Machine and Operator.

Question 9b(ii) (2 marks)

Calculate the utilization of the machines.

Question 9b(iii) (1 mark)

Rafi is asked to find out how much space to allocate for parts waiting for a machine. Which statistic should he look at?

_____

Question 9c (2 marks)

Give 2 reasons why Rafi should build a 3D simulation model instead of a process flow model to study the production line.

Question 10 (14 marks)

Figure 6a shows a simulation model of customer checkout at a mini mart. There are 2 checkout counters – 1 normal checkout and 1 express checkout.

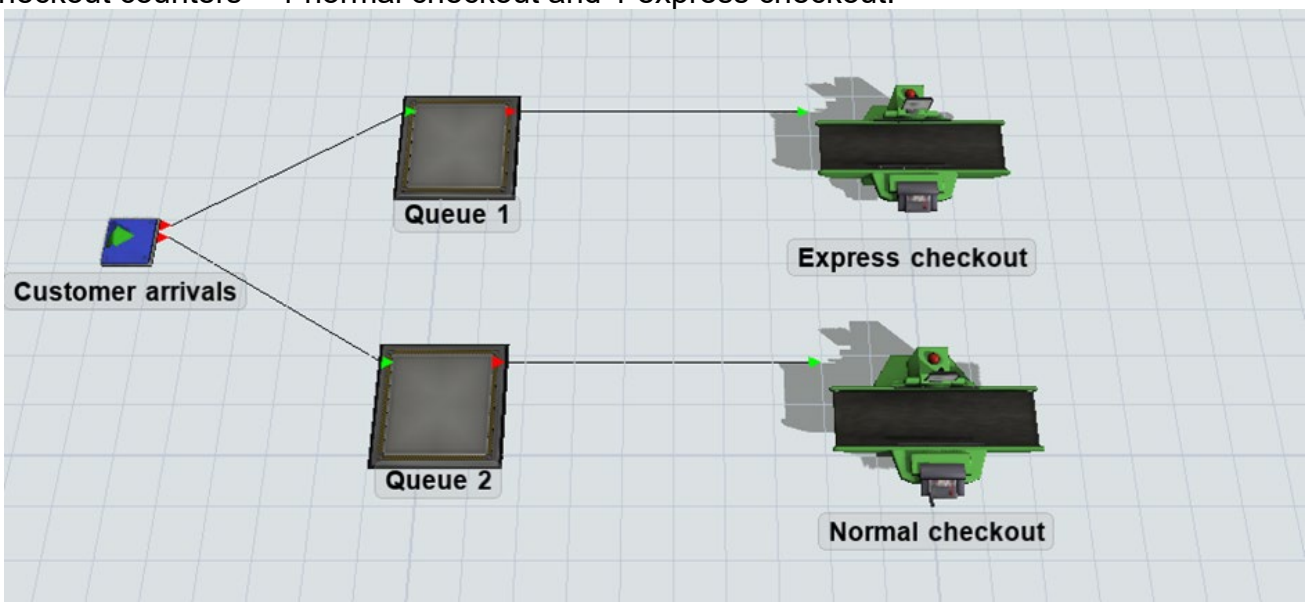

Figure 6a Customer Checkout Model

Question 10a (2 marks)

Referring to Figure 6a, briefly describe what must be done to complete the 3D model.

Question 10b (4 marks)

At the 'Customer arrivals' object, an On Creation trigger was created as shown in Figure 6b. Explain what this trigger does.

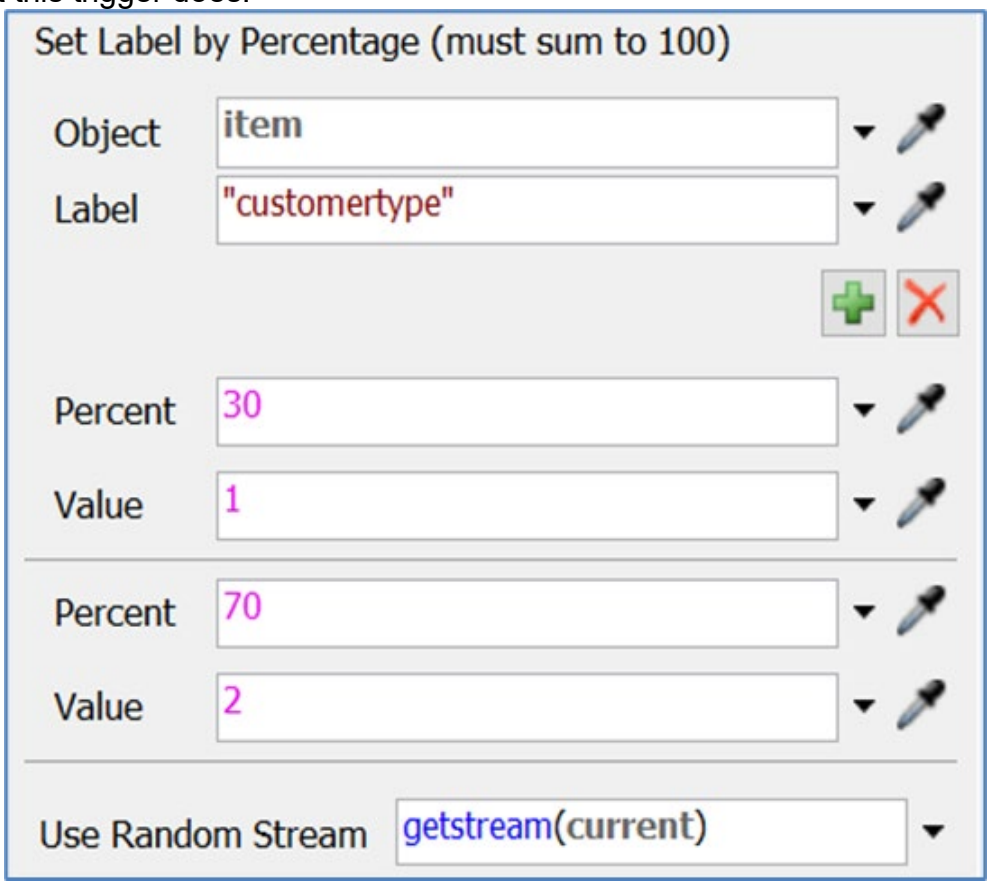

Figure 6b On Creation Trigger

Question 10c (3 marks)

Assume that 30% of customers will use Express checkout. At the Send to Port option of 'Customer arrivals' object, what should be the input for each blank to simulate the flow correctly?

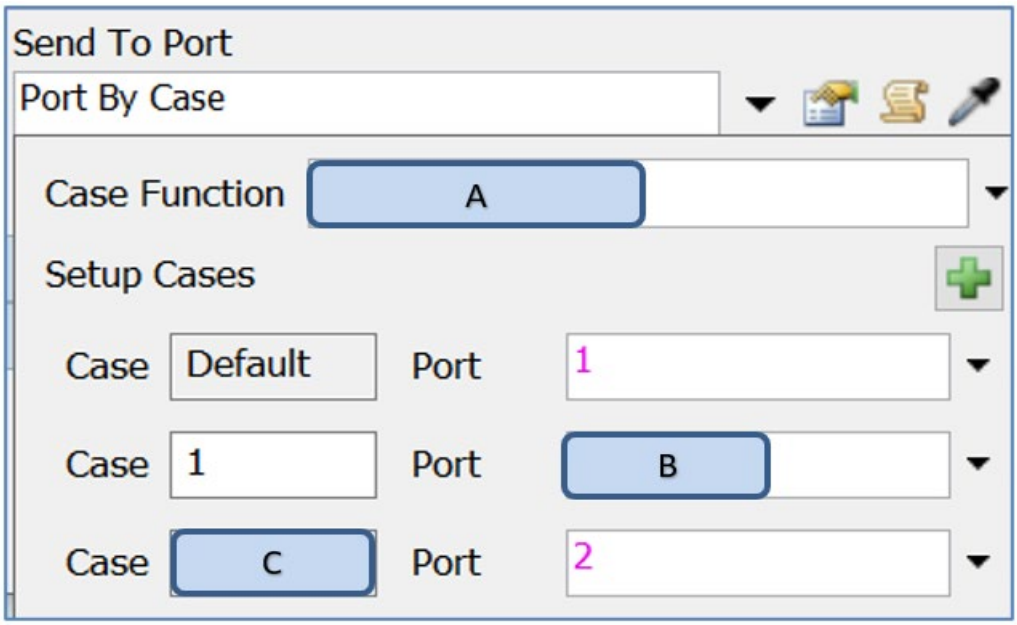

Figure 6c Send to Port Properties of Customer Arrivals

A: _____
B: _____
C: _____

Question 10d (2 marks)

Figure 6d shows the scenario comparison outcome from Experimenter runs. Each scenario has a different number of checkout counters (1, 2, 3 and 4 respectively). Based on the results shown, how many normal checkout counters would you recommend for the mini mart? Briefly explain.

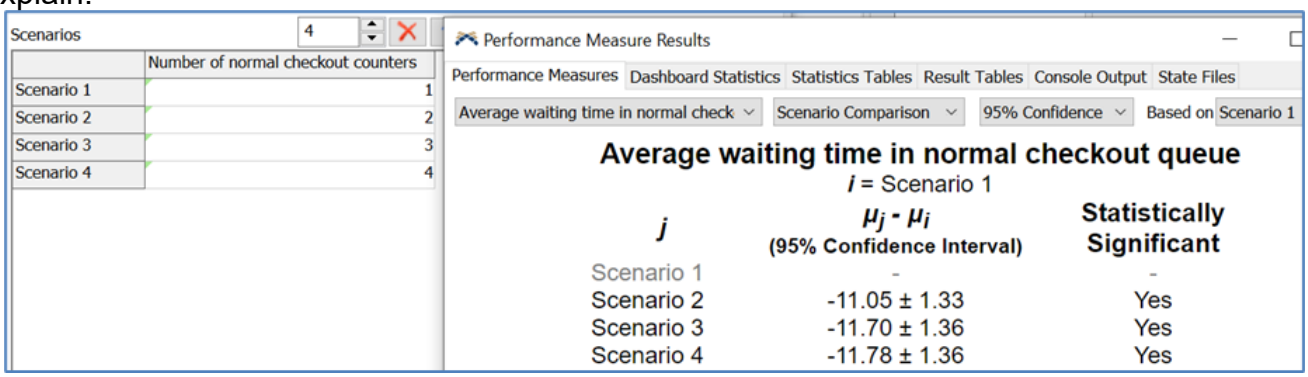

Figure 6d Scenario Comparison Results

Question 10e (3 marks)

State 3 pitfalls to watch out for when conducting a simulation study.

Question 11 (9 marks)

David, the Operations Manager of Greenlife Clinic created a 3D model of the clinic with FlexSim Healthcare Simulation as shown in Figure 7a.

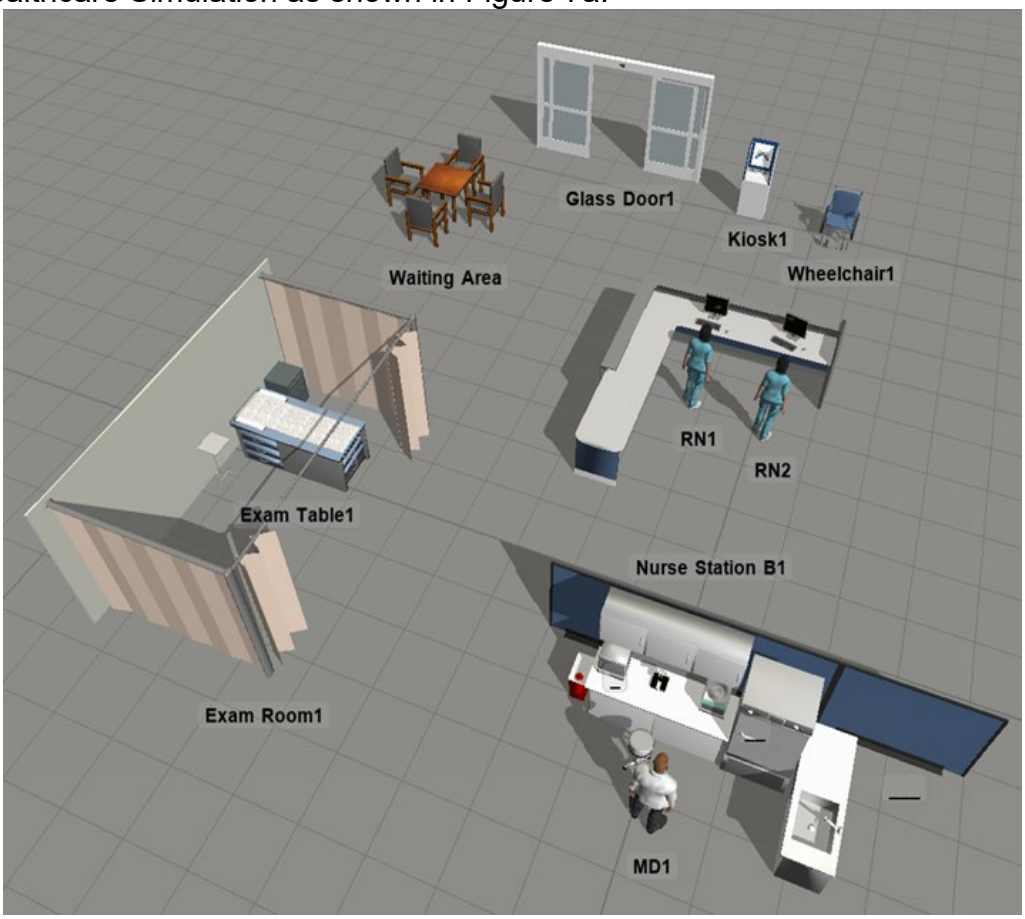

Figure 7a Greenlife Clinic 3D Model

The clinic patient flow is given in Table 7a.

Table 7a Clinic Patient Flow

Step 1. Patients arrive at the glass door.	Step 2. Patients register at Kiosk.	Step 3. Patients wait in the waiting areas for one of the nurses and the examination table to be available.	Step 4. Nurse escorts patients to examination room and prepares for MD examination.	Step 5. MD examines the patients.	Step 6. At the end of MD examination, same nurse escort patient to exit.

Question 11a (1 mark)

Referring to model in Figure 7a, **Kiosk 1** in the model can be found in

the _____ 3D object of the FlexSim Healthcare library.

Question 11b (2 marks)

Given the patient arrival schedule below, where should David enter this data in the FlexSim Healthcare model?

Start (DD:)HH:MM:SS	End (DD:)HH:MM:SS	Quantity	Name
08:00:00	08:30:00	5	
08:30:00	09:30:00	3	
09:30:00	10:00:00	3	

Figure 7b Patient Arrival Schedule

Question 11c (2 marks)

For the flow in Figure 7c, recommend David the appropriate **HC activity set** to simulate Step 2 and Step 4 respectively. Refer to the flow in Table 7a above.

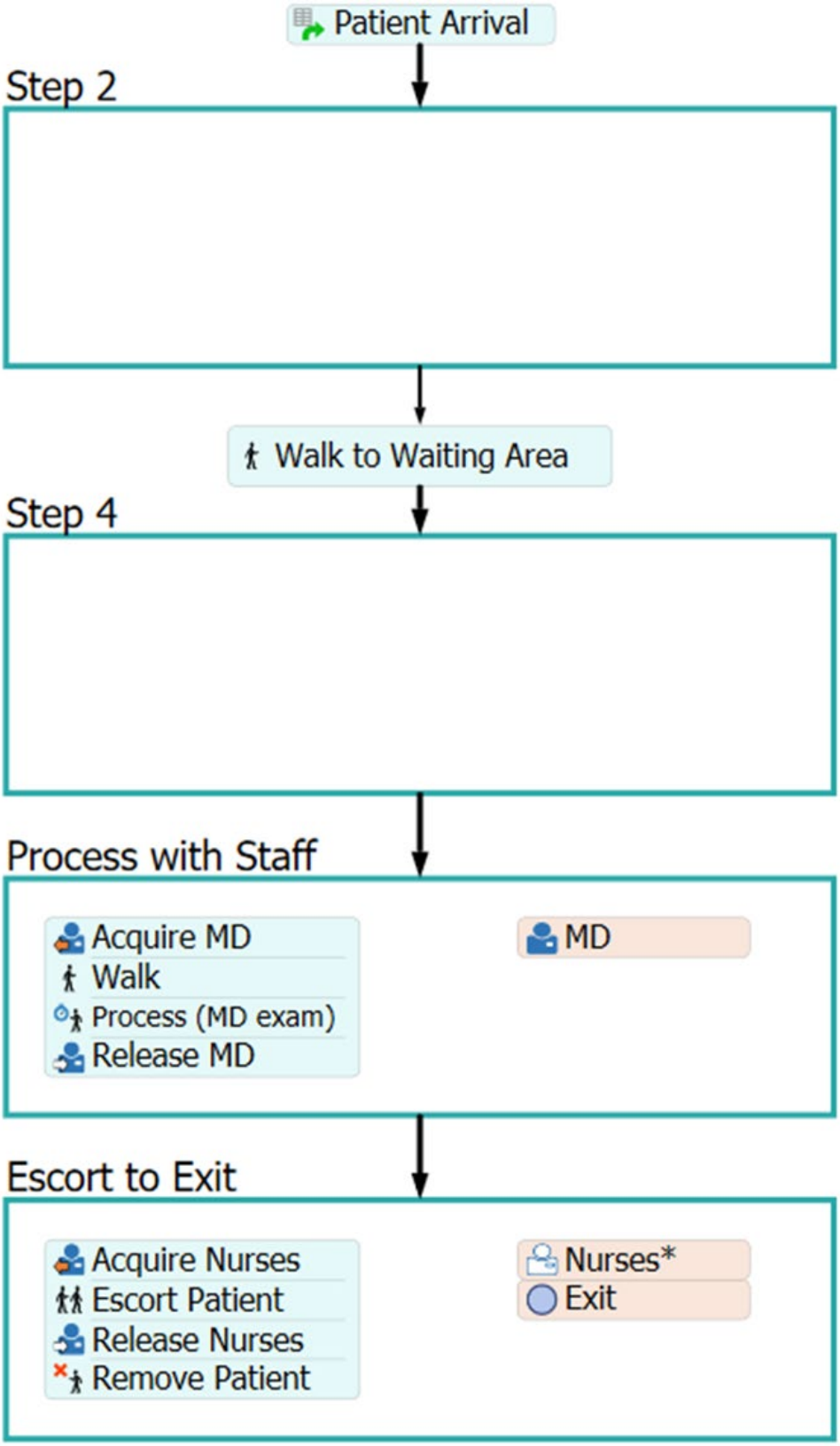

Figure 7c Incomplete Patient Flow in FlexSim

Step 2: _____
Step 4: _____

Question 11d (4 marks)

A new requirement is added for all patients to be transported with wheelchair when they are escorted to the exit. Name the relevant activities from HC Sub Flows and HC Resources in the Library that should be added to the '**Escort to Exit**' activity set in order to simulate the new requirement.

END OF SECTION A

End of Assessment